# LINCURVE PRO

## Instructions For Use

# Table of Contents


1 System overview ..... 4
1.1 System description ..... 4
1.2 Intended use ..... 4
1.3 System components ..... 5
1.4 Technical Specifications ..... 6
2 Safety Precautions ..... 7
2.1 Warnings and Precautions ..... 7
2.2 General precautions ..... 7
2.3 Maintenance precautions ..... 7
2.4 Precautions during use ..... 7
2.5 Safety precautions ..... 8
2.6 Pre-procedure checklist ..... 9
2.7 Procedure precautions ..... 10
2.8 Expected adverse events ..... 11
3 Operating and storage environment ..... 11
4 System components and features ..... 12
4.1 Main body (Body+LCD) ..... 12
4.2 Accessories ..... 13
5 Installation procedure ..... 15
5.1 Pre-installation checklist ..... 15
5.2 Main body (Body & LCD) assembly ..... 15
5.3 Pre-use checklist ..... 16
5.4 Power connection ..... 16
5.5 Handpiece connection ..... 16
5.6 Cartridge connection ..... 17
5.7 Cartridge replacement (Applicable to S type, L type, D type) ..... 17
5.8 Power On/Off ..... 18
6 Operating Instructions ..... 21
6.1 Main screen configuration ..... 21
6.2 Mode description ..... 22
6.3 Info screen configuration ..... 25

6.4 Instructions ..... 26
7 Cleaning and Care ..... 32
7.1 Regular maintenance ..... 32
7.2 Cleaning ..... 32
7.3 Storage ..... 33
8 Troubleshooting ..... 34
8.1 Troubleshooting guidelines ..... 34
8.2 System messages ..... 35
9 IEC Electrical Safety and EMC Specifications ..... 39
10 Symbol Descriptions ..... 42
11 Warranty ..... 44

# 1 System overview

## 1.1 System description

LINCURVE PRO (NP-110) delivers concentrated ultrasound energy to the lower part of the skin through a focused ultrasound transducer. A small thermal coagulation zone is formed in the deepskin layers such as the dermis, the boundary layer between the dermis and subcutaneous fat, the SMAS layer, and the fascia. The area where ultrasound energy is concentrated causes coagulation of the tissue, and gradual thermal damage is created in nearby tissues, leading to various degrees of collagen degeneration. This process stimulates the tissue's natural healing response, resulting in skin lifting, tightening, and improved elasticity.

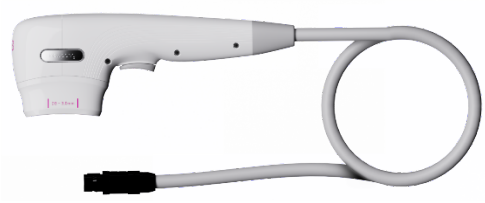
## 1.2 Intended use

The LINCURVE PRO is used to achieve skin lifting and tightening effects by inducing the regeneration of collagen and elastic Fibers. This is accomplished when focused ultrasound energy selectively coagulates the subcutaneous and SMAS layers for the indications for use described below.

- Eyebrow and sagging chin lifting/tightening
- Improves firmness of skin on jaw line, peri-orbital, cheek, abdomen, thigh and arm
- Improve fine lines and wrinkles on the chest(decolletage) and neck area
- Improving cellulite and stretch marks

## 1.3 System components

Main body (Body + LCD)



Handpiece

Cartridge (S2, L2, L3, L4, D3, D4)

Power cord

LCD holder pin

User manual

| Main body | | |
|---|---|---|
| Item | Quantity | Remarks |
| Body | 1ea | |
| LCD | 1ea | |
| **Accessories** | | |
| Item | Quantity | Remarks |
| Handpiece | 1ea | |
| Cartridge S2 | 1ea | Optional item |
| Cartridge L2 | 1ea | Optional item |
| Cartridge L3 | 1ea | Optional item |
| Cartridge L4 | 1ea | Optional item |
| Cartridge D3 | 1ea | Optional item |
| Cartridge D4 | 1ea | Optional item |
| Power cord | 1ea | |
| LCD holder pin | 2ea | |
| User manual | 1ea | |

## 1.4 Technical Specifications

<table>
<tr><th>Category</th><th>Description</th><th>Remarks</th></tr>
<tr><td>Product name</td><td>LINCURVE PRO</td><td></td></tr>
<tr><td>Model number</td><td>NP-110</td><td></td></tr>
<tr><td>Input voltage</td><td>100-240VAC, 50/60Hz</td><td></td></tr>
<tr><td>Power consumption</td><td>120VA</td><td></td></tr>
<tr><td>Fuse</td><td>T6.3A-250V / 2ea</td><td></td></tr>
<tr><td>Output setting and frequency</td><td>
<table>
<tr><th></th><th>DEEP</th><th>BOOSTER</th><th>Frequency</th></tr>
<tr><td>S2</td><td>0.11 ~ 0.25 J</td><td>-</td><td>7MHz</td></tr>
<tr><td>L2</td><td>0.11 ~ 0.25 J</td><td>2.5 ~ 5.5J</td><td>7MHz</td></tr>
<tr><td>L3</td><td>0.20 ~ 0.45 J</td><td>2.5 ~ 5.5J</td><td>7MHz</td></tr>
<tr><td>L4</td><td>0.45 ~ 1.00 J</td><td>4.7 ~ 10.5J</td><td>4MHz</td></tr>
<tr><td>D3</td><td>0.20 ~ 0.45 J</td><td></td><td>7MHz</td></tr>
<tr><td>D4</td><td>0.45 ~ 1.00 J</td><td></td><td>4MHz</td></tr>
</table>
</td><td></td></tr>
<tr><td rowspan="3">Mode setting</td><td>SINGLE mode/AUTO mode</td><td>-</td></tr>
<tr><td>DEEP mode (SINGLE mode /AUTO mode)</td><td>L, D Cartridge</td></tr>
<tr><td>BOOSTER mode (SINGLE mode/AUTO mode)</td><td>Use L Cartridge only</td></tr>
<tr><td>Protection against electric shock</td><td>Class I device</td><td></td></tr>
<tr><td>Degree of patient protection</td><td>Type BF Applied part</td><td></td></tr>
<tr><td>Safety standards</td><td>IEC 60601-1, IEC 60601-1-2</td><td></td></tr>
<tr><td>Ingress protection</td><td>Cartridge (IPX7)<br>(Excluding the top part of the cartridge)</td><td>Main Body<br>& Handpiece is not waterproof.</td></tr>
<tr><td>Temperature stability of contact surfaces</td><td>The temperature of all contactable fittings is within 43°C.</td><td></td></tr>
<tr><td>Physical Dimensions<br>(L x H x W)</td><td>298.8 x 368.1 x 418.9 mm</td><td></td></tr>
<tr><td>Weight</td><td>6.7Kg</td><td></td></tr>
<tr><td>Warranty</td><td>- Main body: 2 years<br>- Accessories (Handpiece, Cartridge):<br>1 year or the specified shot number is used whichever comes first</td><td></td></tr>
</table>

# 2 Safety Precautions

## 2.1 Warnings and Precautions

1) Read and understand the instructions before use.
2) Only the authorized operator may use this device.
3) The procedure must be performed by a doctor or licensed medical personnel under the direction and supervision of a doctor.
4) Dispose of the main unit and cartridge in accordance with applicable regulations. For any issues related to disposal, please contact Newpong Co., Ltd. or your authorized distributor.

## 2.2 General precautions

1) Before use, be sure to read and understand this user manual, and use only the designated components.
2) Doctors and all personnel involved in the procedure must be familiar with this user manual.
3) If any issues occur during use, contact the manufacturer for assistance.
4) Do not attempt to repair or modify the device unless you are a certified Newpong service technician.
5) All maintenance and service must be performed by authorized technicians using approved components.
6) Do not disassemble or modify the product under any circumstances.

## 2.3 Maintenance precautions

1) Clean the handpiece with alcohol to maintain hygiene.
2) Store the cartridge at room temperature to prevent the internal liquid from freezing, which may cause damage. Do not use frozen cartridges.
3) When cleaning the LCD touchscreen, use a soft cloth moistened with alcohol.
4) To clean the surface of the device, wipe it with a soft cloth moistened with alcohol. Avoid allowing any liquid to enter the device.
5) Ensure that the hose connecting the main unit and the handpiece is not bent or twisted.
6) Do not use or store the device in areas with high humidity or poor ventilation. Avoid exposure to direct sunlight or high-temperature environments.

## 2.4 Precautions during use

1) Do not operate the device near flammable or combustible gases or materials.
2) Check the coolant level inside the cartridge before use. Do not operate the device if the coolant is absent or insufficient.

3) Apply ultrasound gel to the treatment area on the patient's skin to enhance ultrasound transmission. Do not use water or general cosmetic products.
4) Only trained operators may use this device.
5) High humidity may cause condensation on components and surfaces, potentially damaging critical parts. Maintain proper humidity levels.
6) Use the device in a well-ventilated area to prevent damage caused by overheating, dust, or mechanical vibration.
7) Do not install the device outdoors or in environments with severe temperature fluctuations. Install it indoors in a well-ventilated area, away from patient traffic.
8) Handle the handpiece and cartridge carefully to prevent them from falling.
9) Do not press or scratch the screen surface with fingernails or sharp objects (e.g., nails, pencils, pens), as this may cause damage.
10) Always keep the handpiece connector clean and dry.
11) Take care not to drop the handpiece when transporting the device.

## 2.5 Safety precautions

1) Do not place flammable materials such as volatile anesthetics or solvents near the device.
2) Do not use the device in close proximity to other equipment.
3) Do not connect the power cord if it is damaged.
4) To avoid potential electromagnetic hazards or interference between devices, connect only to a power supply with protective grounding. Do not use a multi-socket outlet (MSO).
5) This product complies with IEC 60601-1-2 (EMC) and does not interfere with nearby devices. However, to minimize potential risks, do not place it directly adjacent to or stacked with other electronic equipment.
6) Due to the electrical nature of the device, internal sparking could lead to gas explosions in the presence of flammable gases.
7) Do not place power cables near heating equipment, as this could cause wire breakage or short circuits.
8) Ensure the power supply voltage, frequency, and current (or power consumption) match the device's specifications.
9) Make sure the power plug is securely inserted into the outlet.
10) To prevent electric shock, connect the device only to a power supply with protective grounding.
11) Use only the dedicated power cord provided by the manufacturer. Use of unverified components may result in malfunction or electric shock.
12) All cables must be connected correctly and securely.
13) The device must be grounded. When not in use, always switch off the main power.

14) Do not use damaged wires, as this could result in fire due to electrical leakage.
15) Do not touch the power plug with wet hands, as this may cause electric shock.
16) Do not insert multiple plugs into a single outlet, as this may lead to fire hazards.
17) Avoid tilting, vibration, or mechanical shock during transport.
18) Be especially cautious when moving the product, as significant damage may occur.
19) Do not pour water on the device or place heavy objects on it.
20) Keep a fire extinguisher near the device in case of fire caused by electrical leakage.
21) Do not place reflective or flammable objects near the device.
22) If a malfunction occurs and a qualified technician is not available, attach a warning label to the device and contact the manufacturer for service.
23) Immediately contact the manufacturer for inspection if any of the following occurs:
    - Abnormal overheating of the main unit
    - Unusual noises during operation
    - Visible deformation or cracking of the main unit, handpiece, cartridge, or other components
24) In hospital environments, hazardous currents may occur due to potential differences between conductive parts. To prevent this, connect the device's equipotential grounding terminal to the hospital's equipotential grounding system.
25) During procedures, bubbles may form on the surface of the transducer, at the boundary between the transducer and tissue, or in the treatment area. To prevent this, ensure the cartridge's film surface is always facing downward during ultrasound delivery.
26) Handle the handpiece with an attached cartridge carefully to avoid any negative impact on the device.
27) If the device is subjected to a strong impact, a full performance inspection must be conducted to ensure all essential functions are operating properly.

## 2.6 Pre-procedure checklist

1) Do not use the device on patients who meet any of the following conditions:
    - Patients with skin allergies to local anesthetics
    - Pregnant women or infants
    - Patients with keloid-prone skin
    - Patients with pacemakers, stents, or metal implants
    - Patients with skin cancer
2) Patients with the following conditions must consult with a medical specialist before proceeding with the treatment:
    - Pregnant or breastfeeding women

- (For women with suspected pregnancy, a pregnancy test must be performed and a consultation with a specialist is required before treatment.)
- Patients taking aspirin or aspirin-based medications
- Patients with bleeding disorders, thrombosis, or other blood coagulation issues
- Patients with cardiovascular or other systemic diseases
- Patients currently suffering from acute or chronic illnesses, including dermatologic conditions
- Patients with dermatologic lesions or abnormal sensation in the treatment area
- Patients with a history of skin resurfacing treatments in the treatment area
- Patients with photosensitive disorders or those taking photosensitizing medications
- Individuals prone to keloids or hypertrophic scarring
- Patients undergoing long-term steroid or immunosuppressive therapy
- Elderly patients aged 65 or older
- Patients with heart disease
- Patients with herpes virus infection
- Patients with diabetes or epilepsy

3) The focused ultrasound energy emitted by this device may be affected by or cause damage to certain materials. Therefore, do not use the device on patients who have any of the following:
   - Mechanical implants
   - Dermal fillers
   - Breast implants

## 2.7 Procedure precautions

1) Apply ultrasound gel to ensure close contact between the cartridge and the treatment area during the procedure. Failure to maintain proper contact may result in inaccurate focal depth, reduced treatment efficacy, or skin burns. Be careful not to allow air bubbles to form within the gel, as they may cause thermal damage to surrounding tissues.
2) There is a risk of bubble formation along the acoustic path during treatment, particularly at the interface between the transducer and tissue. To avoid this, ensure the film surface of the cartridge is always facing downward during ultrasound emission. Additionally, air bubbles in the acoustic path can reduce treatment effectiveness. Therefore, only degassed water should be filled in the cartridge.
3) The uniform propagation of High-Intensity Focused Ultrasound (HIFU) waves may be disrupted by bone or gas-containing organs in the treatment area. Use caution when treating areas with loose or sagging skin, as there is a risk of facial nerve injury.

4) Patients may experience pain or itching during the procedure. If such symptoms occur, follow the physician's instructions.
5) The following temporary side effects may occur after treatment. In case of side effects, follow the instructions of a medical professional:
   - Numbness
   - Stiffness
   - Nerve paralysis
   - Erythema, itching, or swelling
   - Pain, bruising, erosion, blistering, or ulceration
   - Hypopigmentation or hyperpigmentation
6) If abnormal symptoms such as skin depressions or excessive swelling occur, stop the treatment immediately.
7) Do not apply treatment repeatedly to the same area in succession.
8) Before treatment, clearly explain all possible side effects to the patient and obtain a signed informed consent form.
9) When using the S2 cartridge, which emits single shots, avoid overlapping exposures on the same spot. Move continuously during treatment to prevent damage.

### 2.8 Expected adverse events

1) Unintended heating of tissues may occur, potentially resulting in skin damage or fistula formation.
2) Thermal injury may affect the skin, surrounding tissues, or adjacent anatomical structures.

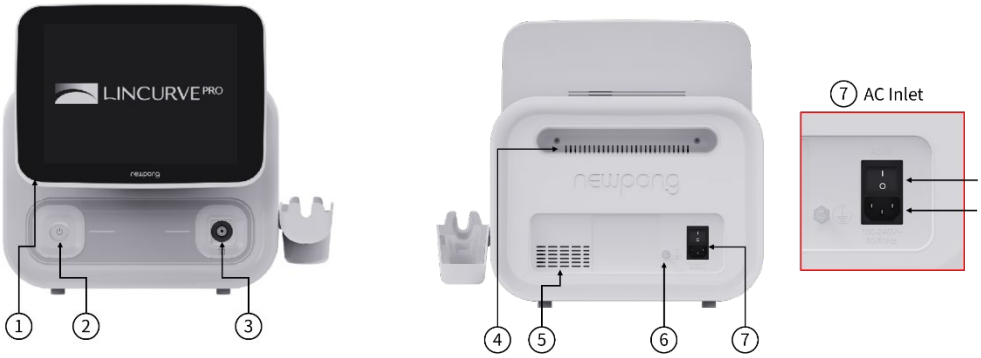
## 3 Operating and storage environment

| Condition | Operating | Storage |
|---|---|---|
| Temperature | 5 ~ 35°C | Main body: -10 ~ 50°C<br>Cartridge: 2 ~ 50°C |
| Humidity | 20 ~ 90 % | 20 ~ 90 % |
| Atmosphere | 900hPa ~ 1060hPa | 900hPa ~ 1060hPa |
| Remarks | | Store in a dry place at room temperature, away from direct sunlight. |

# 4 System components and features

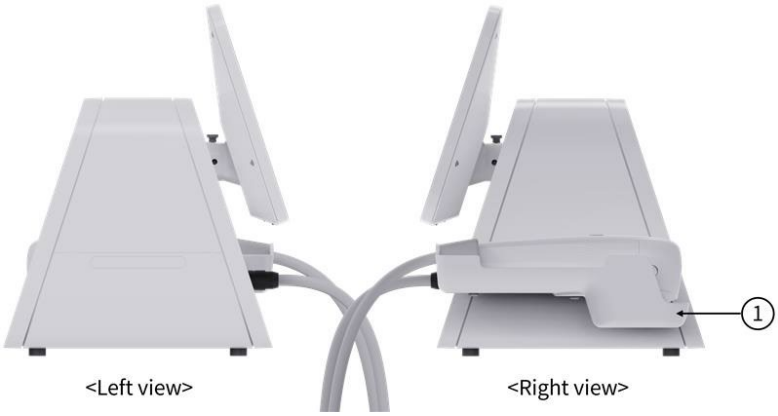
## 4.1 Main body (Body+LCD)

### 4.1.1 Front and rear view




| # | Item | Description |
|---|---|---|
| ① | TFT LCD | Various user interface (UI) elements are displayed |
| ② | auxiliary power button | Turn system On/Off |
| ③ | Handpiece connector | Socket for plugging in handpiece cable |
| ④, ⑤ | Fan | Air circulation for cooling |
| ⑥ | Protective earth | protective earth(ground) symbol and pin |
| ⑦ | AC Inlet | Supplies power to system |

### 4.1.2 Left and right view




| # | Item | Description |
|---|---|---|
| ① | Handpiece Holder | Place to put the handpiece |

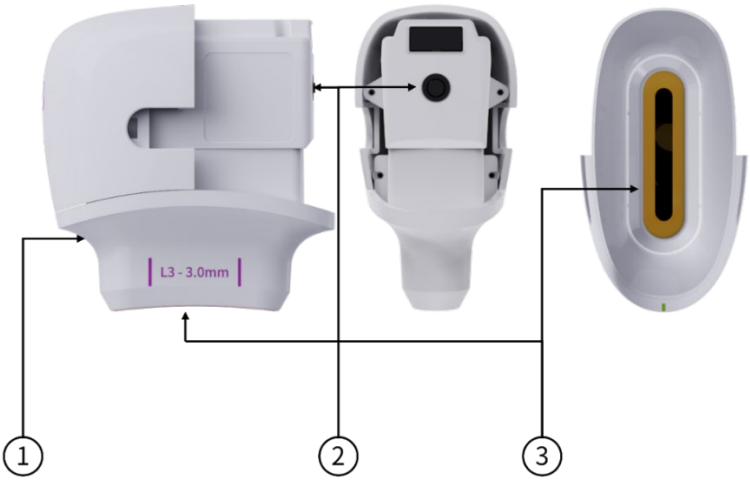
## 4.2 Accessories

### 4.2.1 Handpiece

<Cartridge connection>

| # | Item | Description |
|---|---|---|
| ① | Handpiece plug | Connect to the main body's handpiece cable connector |
| ② | Handpiece cable | Cable for the handpiece |
| ③ | Handpiece body | Handpiece Body |
| ④ | Handpiece shot button | Button to start and stop shot |
| ⑤ | Cartridge release button | Press when removing cartridges |
| ⑥ | LED color | Status: STANDBY — Color: White<br>Status: READY — Color: Green<br>Status: SHOT — Color: Blue |

### 4.2.2 Cartridge (S2, L2, L3, L4, D3, D4)




| # | Item | | Description |
|---|---|---|---|
| ① | Cartridge | Contact cover | Protects the contact film and indicates the output area. |
| ② | | Handpiece Joint | Part that combines with the handpiece. Contains an ultrasonic ceramic and a cartridge board that stores cartridge information and the number of shots. |
| ③ | | Contact Film | Part where focused ultrasound is output and used in close contact with the skin. |

### 4.2.3 LCD holder pin

Pin Provided for Assembling the LCD to the Body

※ Be sure to use only the pins supplied with the device.

# 5 Installation procedure

## 5.1 Pre-installation checklist

1) Before using the device, check the main unit and accessories for any damage or defects.
2) Ensure the cartridge has been properly cleaned before use. Do not operate the device if any accessories are damaged.

## 5.2 Main body (Body & LCD) assembly

1) Properly attach the LCD to the main body.

2) After attachment, insert the LCD holder pin into the corresponding hole.

3) Once the LCD holder pin is secured, the NP-110 main body assembly is complete.

## 5.3 Pre-use checklist

1) Use the device on a flat surface that does not transmit vibrations to the equipment.
2) The device may generate some heat; therefore, ensure that the exhaust vents are not blocked and operate the device in a well-ventilated area.

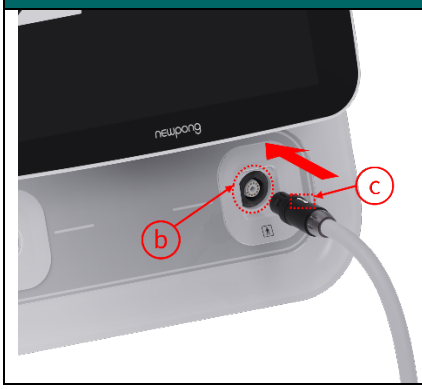
## 5.4 Power connection

Turn off the power switch on the back of the device (position O indicates OFF), then connect the power cord to the power inlet (ⓐ) as shown in the illustration below.

| Before connecting | After connecting |
|---|---|
| | |

## 5.5 Handpiece connection

Align the arrow on the handpiece plug (ⓒ) to face upward toward the handpiece cable connector (ⓑ) and connect it to the handpiece cable connector (ⓑ) located on the front of the device, as shown in the illustration below.

| Before connecting | After connecting |
|---|---|
|   | |

Ensure that the handpiece cable is fully inserted by confirming an audible 'click' sound upon connection.

※ The handpiece must be connected or disconnected only when the device is powered off.

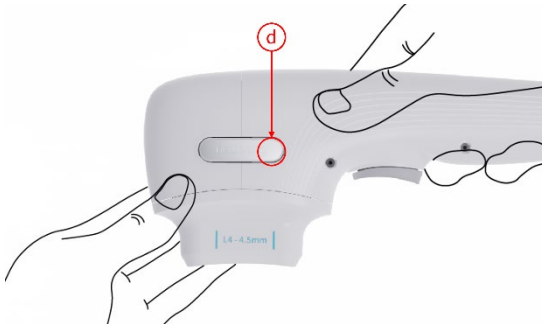
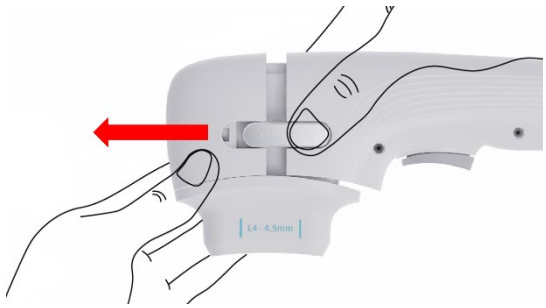
## 5.6 Cartridge connection

After selecting one of the six types of cartridges, insert it into the handpiece until it is fully seated, as shown in the image below.

| Before connecting | After connecting |
| --- | --- |
| | |

## 5.7 Cartridge replacement (Applicable to S type, L type, D type)

Press the cartridge release switch (ⓓ) and pull in the direction of the arrow to remove the cartridge.

| Before removal | Cartridge-Handpiece Disconnection |
| --- | --- |
|  |  |

| After removal |
| --- |
| |

## 5.8 Power On/Off

### 5.8.1 Power on

1) Turn on the power using the "I" mark on the power switch located at the back of the device.

※ When the power switch is in the "O" position, the device is OFF; when in the "I" position, the device is ON.

| Power Off | Power On |
|---|---|
| | |

2) Press the auxiliary power button (ⓔ) on the front of the device to turn on the LCD and activate the device for use.

| Auxiliary power off | Auxiliary power on |
|---|---|
| | |

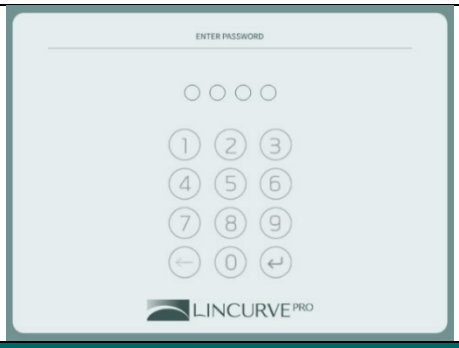
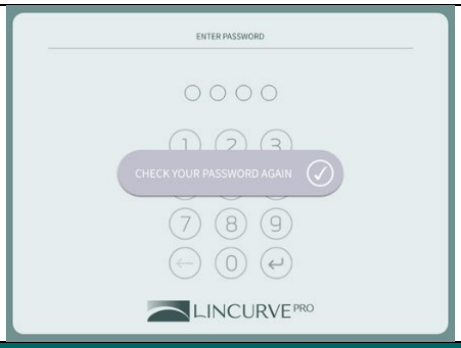
3) When the power is turned on without any issues, the system will boot up, and an intro screen will appear for about 1 second as shown below.

<Intro screen>

4) After the intro screen is displayed, the device will automatically switch to the password entry screen.

5) Enter the 4-digit password provided by the manufacturer, then press the '↵' button to proceed to the main screen.

※ If the password is incorrect, an error message will be displayed for 1 second, and the entered password will be reset.

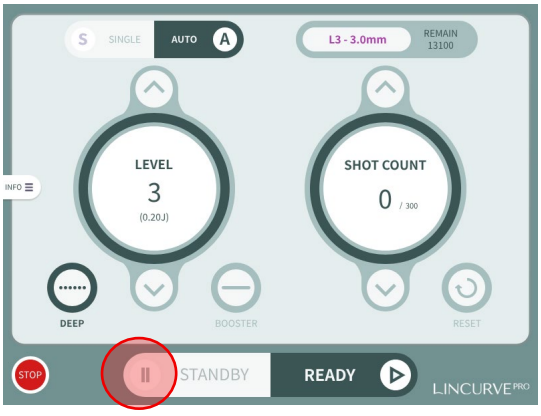
| | |
|---|---|
|  |  |
| <Password input screen> | <Password error message> |

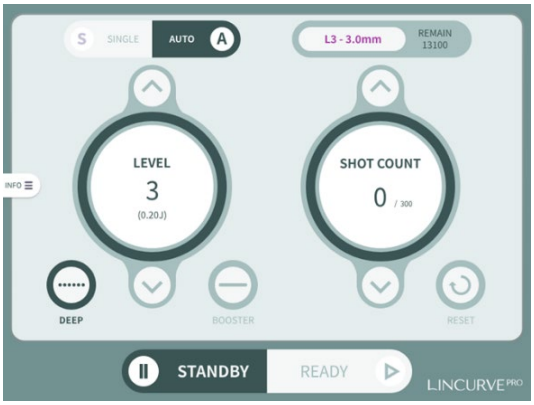




5.8.2 Power Off

1) Press the 'STANDBY' button to switch the STANDBY status.




2) Press and hold the auxiliary power button (ⓔ) on the front of the device for more than 1 second to turn off the LCD and the device.

| Auxiliary power off | Auxiliary power on |
|---|---|
| | |

※ If the auxiliary power button is pressed during a cartridge shot in READY status, the device will finish the current shot before shutting down.

3) Turn off the device by pressing the ‘O’ (OFF) side of the power switch located on the rear panel.

※ The power switch is in the OFF position when set to 'O', and in the ON position when set to 'I'.

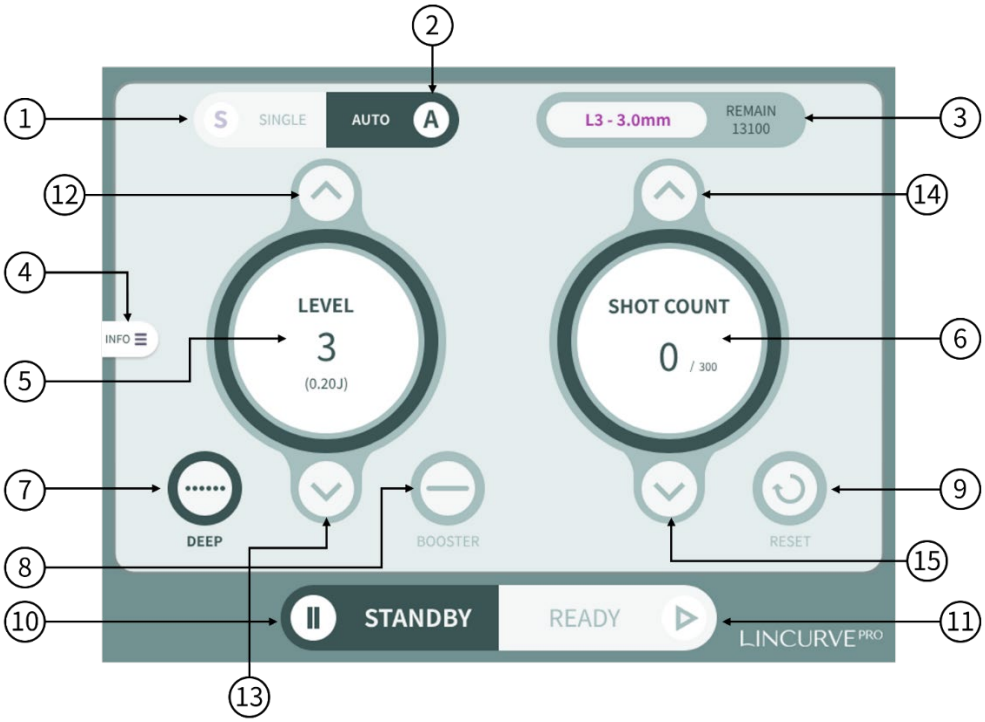
| Main power off | Main power on |
| --- | --- |
| | |

4) After removing the cartridge, clean it thoroughly and store it in a clean, dry place.

5) If the device will not be used for an extended period, switch off the main power located on the rear panel and disconnect the power cable before storing the device.

# 6 Operating Instructions

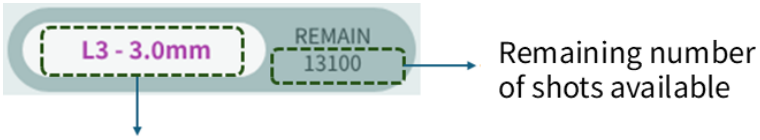
## 6.1 Main screen configuration




① SINGLE: Switches the operating mode to 'SINGLE' mode

② AUTO: Switches the operating mode to 'AUTO' mode

③ Cartridge Information: Displays the type of the connected cartridge and the remaining number of shots available.




| Cartridge | Information | Color |
|---|---|---|
| S2 | S2 – 2.0mm | Yellow |
| L2 | L2 – 2.0mm | Green |
| L3 | L3 – 3.0mm | Purple |
| L4 | L4 – 4.5mm | Sky blue |
| D3 | D3 – 3.0mm | Magenta |
| D4 | D4 – 4.5mm | Blue |

<Connected cartridge type>

④ INFO Button: Switches to the Info screen. (Used to check device information, adjust brightness and volume levels, and configure settings.)

⑤ Output Value: Displays the selected ultrasound energy output level for the procedure.

⑥ Shot Count: In SINGLE mode, displays the number of shots that have been delivered so far.

⑦ DEEP: Switches the output mode to Deep mode.

⑧ BOOSTER: Switches the output mode to Booster mode.

⑨ RESET: Resets the current shot count to zero.

⑩ STANDBY: Sets the STANDBY status.

⑪ READY: Sets the READY status.

⑫ Output Increase Button: Increases the ultrasound energy output level by one step during treatment.

⑬ Output Decrease Button: Decreases the ultrasound energy output level by one step during treatment.

⑭ Shot Count Increase Button: Increases the desired shot count by the configured unit value during treatment (not available in SINGLE mode).

⑮ Shot Count Decrease Button: Decreases the desired shot count by the configured unit value during treatment (not available in SINGLE mode).

## 6.2 Mode description

1) Operating mode and Output mode

| MODE | | Description |
|---|---|---|
| Operating Mode | SINGLE | In SINGLE mode, pressing the handpiece shot button once delivers a single ultrasound output.<br>(Applicable cartridges: S2, L2, L3, L4, D3, D4) |
| | AUTO | - In AUTO mode, pressing the handpiece shot button once delivers continuous ultrasound pulses up to the user-defined shot count.<br>- If the shot button is pressed again before reaching the set shot count, the operation is paused.<br>- Pressing the shot button once more during the pause will resume ultrasound delivery until the remaining shots are completed.<br>(Applicable cartridges: S2, L2, L3, L4, D3, D4) |
| Output Mode | BOOSTER | A mode in which ultrasound energy is delivered continuously.<br>Single Booster output: <br>(Applicable cartridges: L2, L3, L4) |
| | DEEP | A mode in which ultrasound energy is delivered at regular intervals.<br>Single Deep output: <br>(Applicable cartridges: L2, L3, L4, D3, D4) |

※ SINGLE mode is the default operation mode, and DEEP mode is the default output mode.

2) Operating Modes by Cartridge Type

| Cartridge | Mode | | Description | Remarks |
|---|---|---|---|---|
| S Type | SINGLE | | Pressing the handpiece shot button once delivers a single ultrasound pulse. | - LED lights blue during ultrasound output.<br>- DEEP mode and BOOSTER mode are not available for S-type cartridges.<br>- In AUTO mode, the shot count can be set from 50 to 950. |
| | AUTO | | Delivers the preset number of shots with one button press. | |
| L Type | SINGLE | DEEP | Single Button-Press Output | - LED lights blue during ultrasound output. |
| | | BOOSTER | Single Button-Press Output | |
| | AUTO | DEEP | Delivers the preset number of shots with one button press. | - LED lights blue during ultrasound output.<br>- In AUTO mode, the shot count can be set from 50 to 950. |
| | | BOOSTER | Delivers the preset number of shots with one button press. | |
| D Type | SINGLE | DEEP | Single Button-Press Output | - LED lights blue during ultrasound output. |
| | AUTO | DEEP | Delivers the preset number of shots with one button press. | - LED lights blue during ultrasound output.<br>- In AUTO mode, the shot count can be set from 50 to 950. |

3) On the main screen, select the desired mode, then choose the power level and shot count.

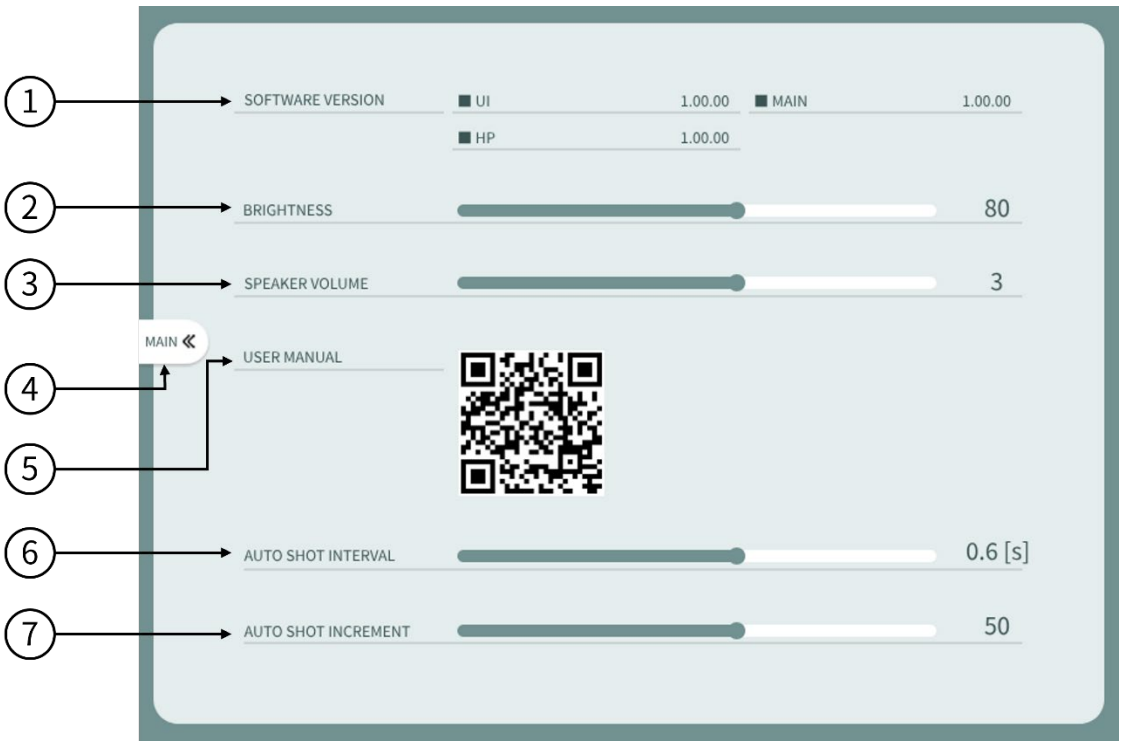
① Power setting

② Shot count selection is available only in AUTO mode. The shot count can be set between 50 and 950 using the ^/v buttons.

| S type cartridge | | | | | |
|---|---|---|---|---|---|
| Cartridge | Mode | | Min. Power(J) | Max. Power(J) | Remarks |
| S2 2.0mm | SINGLE | | 0.11 | 0.25 | * Selectable levels from 3 to 10.<br>* Use the ^/v buttons to adjust the output level in increments of 1. |
| | AUTO | | | | |
| **L type cartridge** | | | | | |
| Cartridge | Mode | | Min. Power(J) | Max. Power(J) | Remarks |
| L2 2.0mm | DEEP | SINGLE | 0.11 | 0.25 | * Selectable levels from 3 to 10.<br>* Use the ^/v buttons to adjust the output level in increments of 1.<br>* In AUTO mode, the shot count can be set from 50 to 950. |
| | | AUTO | | | |
| | BOOSTER | SINGLE | 2.5 | 5.5 | |
| | | AUTO | | | |
| L3 3.0mm | DEEP | SINGLE | 0.20 | 0.45 | |
| | | AUTO | | | |
| | BOOSTER | SINGLE | 2.5 | 5.5 | |
| | | AUTO | | | |
| L4 4.5mm | DEEP | SINGLE | 0.45 | 1.00 | |
| | | AUTO | | | |
| | BOOSTER | SINGLE | 4.7 | 10.5 | |
| | | AUTO | | | |
| **D type cartridge** | | | | | |
| Cartridge | Mode | | Min. Power(J) | Max. Power(J) | Remarks |
| D3 3.0mm | DEEP | SINGLE | 0.20 | 0.45 | * Selectable levels from 3 to 10.<br>* Use the ^/v buttons to adjust the output level in increments of 1.<br>* In AUTO mode, the shot count can be set from 50 to 950. |
| | | AUTO | | | |
| D4 4.5mm | DEEP | SINGLE | 0.45 | 1.00 | |
| | | AUTO | | | |

## 6.3 Info screen configuration

1) Pressing the INFO button displays the screen shown below.




<Info screen>

① SOFTWARE VERSION: Displays the current software version installed on the device.

② BRIGHTNESS: Adjusts the screen brightness. (Range: 50% to 100%, adjustable in 10% increments; Default: 80%)

③ SPEAKER VOLUME: Adjusts the device's speaker volume. (Range: 0 to 5, adjustable in steps of 1; Default: 3)

④ MAIN: Returns to the main screen.

⑤ USER MANUAL: Access the user manual by scanning the QR code.

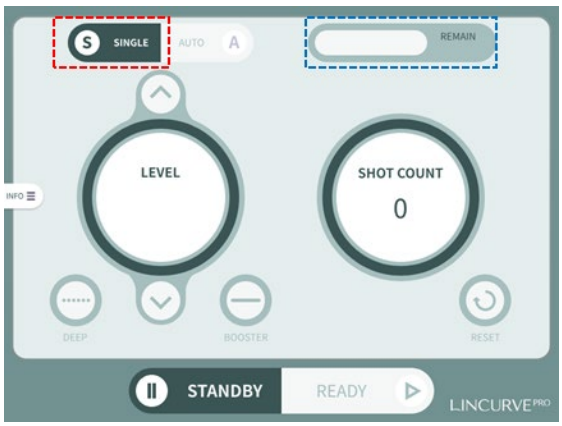
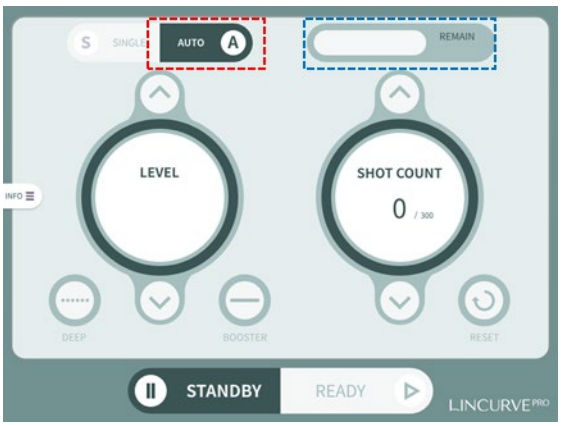
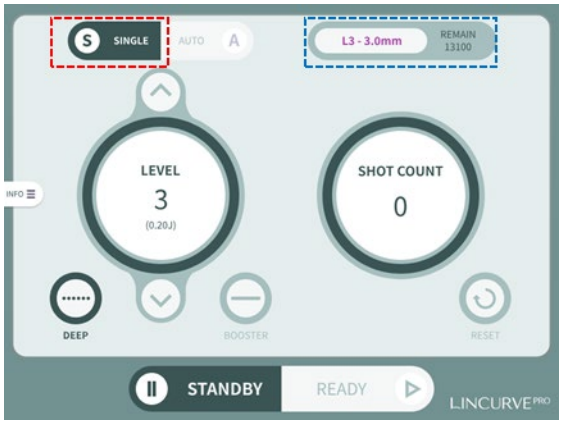
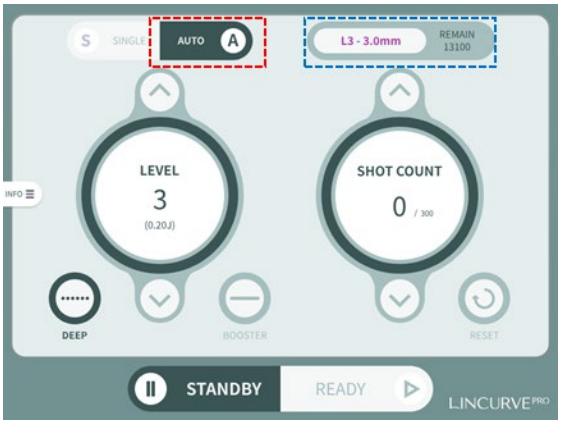
⑥ AUTO SHOT INTERVAL: Sets the time interval between automatic shots. (Range: 0.1 to 1 second, adjustable in 0.1-second increments; Default: 0.1 sec)

⑦ AUTO SHOT INCREMENT: Sets the unit value for automatic shot count adjustments. (Selectable: 10, 50, or 100 shots; Default: 50)

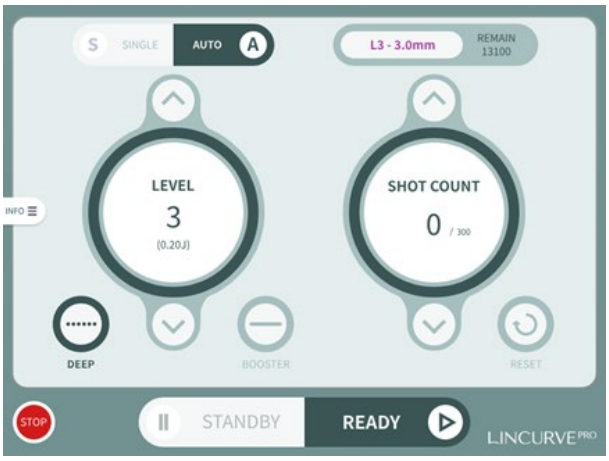
## 6.4 Instructions

### 6.4.1 STANDBY status

1) Pressing the STANDBY button activates the STANDBY status, allowing you to configure the output level, mode, and shot count.
2) In STANDBY status, pressing the handpiece shot button does not trigger any output.
3) While in STANDBY status, the LED on the top of the handpiece lights white.
4) The display screen layout varies depending on whether SINGLE or AUTO mode is selected.
5) If no cartridge is connected, cartridge information (refer to the blue box) cannot be displayed. (The device cannot enter READY status without a connected cartridge.)

| | SINGLE mode - STANDBY screen | AUTO mode - STANDBY screen |
|---|---|---|
| Cartrid |  |  |
| Cartrid |  |  |

### 6.4.2 READY status




1) After completing the settings, press the 'READY' button to enter the READY status.
2) The 'STOP' button is only enabled in the READY status, and all other buttons (except STANDBY/STOP) are disabled.
   ※ The 'STOP' button is used for emergency shutdown. If the 'STOP' button is pressed during output, the output will immediately stop and the device will return to STANDBY status.
3) Output will only begin when the shot button on the handpiece is pressed.
4) During output, the LED on the top of the handpiece will display the preset color according to the selected mode.

| | Color | Photo |
|---|---|---|
| STANDBY | White | |
| READY | Green | |
| Output | Blue | |

※ If the user switches from READY to STANDBY status while the handpiece is in operation, the device will complete the current output and then switch to STANDBY status.
※ If the cartridge is removed or a caution condition occurs, a warning message will appear. After confirming the message, the system will automatically switch to STANDBY status.
※ If the user experiences a sensation of excessive heat or electric shock during operation, immediately remove the handpiece from the patient and press the STOP or STANDBY button to halt the operation. Then, turn off the power and contact the manufacturer or authorized distributor.

### 6.4.3 SINGLE mode

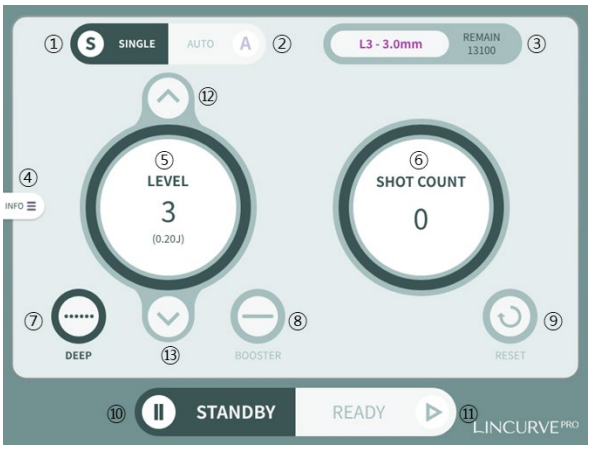
In SINGLE mode, one press of the handpiece shot button delivers a single ultrasound output.

#### 6.4.3.1 S type, D type cartridge

| <S type> | <D type> |
|---|---|

- In STANDBY status, press Button ① to select SINGLE mode.
- Check the connected cartridge information in Section ③ to ensure there are no issues.
- Use Button ⑩ or ⑪ to set the desired output level for the procedure.
- After completing the desired settings, press Button ⑨ to switch to READY status.
- Each time the handpiece shot button is pressed, a single ultrasound output is delivered.
- The number of shots delivered can be monitored in Section ⑥.
- Pressing Button ⑧ stops the output and returns the device to STANDBY status.

#### 6.4.3.2 L type cartridge




<L type>

1) DEEP mode

- In STANDBY status, press Buttons ① and ⑦ to select SINGLE - DEEP mode.
- Verify that the connected cartridge information in Section ③ is normal and free of errors.
- Use Buttons ⑫ or ⑬ to set the desired output level for the procedure.
- Once all settings are configured, press Button ⑪ to switch to READY status.
- Each press of the handpiece shot button delivers a single ultrasound output.
- After each output, the total number of shots delivered can be checked in Section ⑥.
- To stop the output and return to STANDBY status, press Button ⑩.

2) BOOSTER mode

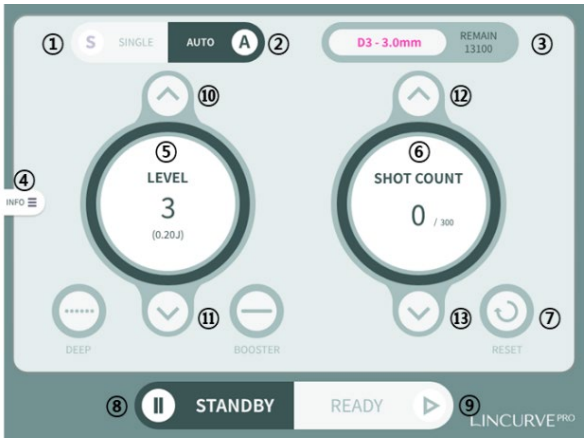
- In STANDBY status, press Buttons ① and ⑧ to select SINGLE - BOOSTER mode.
- Check the connected cartridge information in Section ③ to ensure there are no issues.
- Use Buttons ⑫ or ⑬ to set the desired output level for the procedure.
- After completing the settings, press Button ⑪ to switch to READY status.
- Each time the handpiece shot button is pressed, a single ultrasound output is delivered.
- After each output, the current shot count can be viewed in Section ⑥.
- Press Button ⑩ to stop the output and return to STANDBY status.

### 6.4.4 AUTO mode

In AUTO mode, pressing the handpiece shot button once delivers continuous ultrasound output up to the preset shot count.

#### 6.4.4.1 D type cartridge

※ For D-type cartridges, only DEEP mode is supported; BOOSTER mode is not available.



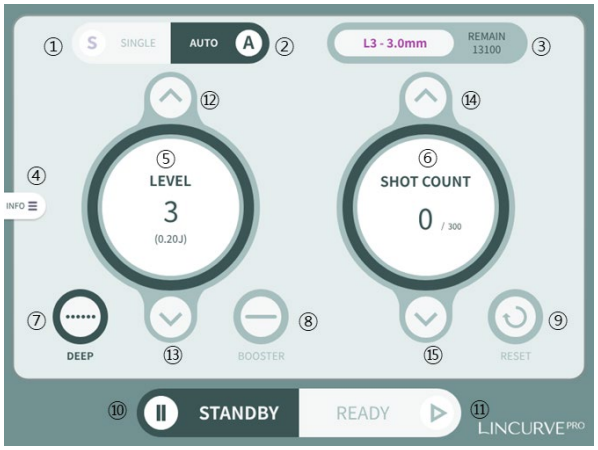

<D type>

- In STANDBY status, press Button ② to select AUTO mode.
- Verify that the connected cartridge information in Section ③ is normal and without

issues.

- Use Buttons ⑩ or ⑪ to set the desired output level for the procedure.
- Use Buttons ⑫ or ⑬ to set the desired shot count.
  (The adjustable shot count range is from 50 to 950.)
- After completing the settings, press Button ⑨ to switch to READY status.
- Pressing the handpiece shot button initiates output up to the specified shot count.
- The current number of shots delivered can be monitored in Section ⑥.
- To pause output before reaching the preset shot count, press the handpiece shot button once more (pause state).
- In pause state, press Button ⑧ to return to STANDBY status.
- Upon reaching the preset shot count, the device resets to SINGLE mode and enters STANDBY status.

### 6.4.4.2 L type cartridge




<L type>

1) DEEP mode

- In STANDBY status, press Buttons ② and ⑦ to select AUTO - DEEP mode.
- Verify that the connected cartridge information displayed in Section ③ is correct and error-free.
- Use Buttons ⑫ or ⑬ to set the desired output level for the procedure.
- Use Buttons ⑭ or ⑮ to set the desired shot count.
  (The adjustable shot count range is from 50 to 950.)
- After configuring the desired settings, press Button ⑪ to switch to READY status.
- Pressing the handpiece shot button initiates continuous output up to the preset shot count.

- The number of shots delivered can be monitored in Section ⑥.
- To pause output before the preset shot count is reached, press the handpiece shot button once again (pause state).
- While in pause state, press Button ⑧ to return to STANDBY status.
- Once the preset shot count is reached, the system automatically resets to SINGLE - DEEP mode in STANDBY status.

2) BOOSTER mode

- In STANDBY status, press Buttons ② and ⑧ to select AUTO - BOOSTER mode.
- Verify that the connected cartridge information in Section ③ is normal and error-free.
- Use Buttons ⑫ or ⑬ to set the desired output level for the procedure.
- Use Buttons ⑭ or ⑮ to set the desired shot count.
  (The adjustable shot count range is from 50 to 950.)
- After completing the desired settings, press Button ⑪ to switch to READY status.
- Pressing the handpiece shot button will initiate continuous output up to the preset shot count.
- Each time output is delivered, the current shot count can be monitored in Section ⑥.
- To pause the output before reaching the preset shot count, press the handpiece shot button once again (pause state).
- While in pause state, press Button ⑩ to return to STANDBY status.
- Once the preset shot count is reached, the system automatically resets to SINGLE - DEEP mode in STANDBY status.

# 7 Cleaning and Care

## 7.1 Regular maintenance

Inspect the product according to the intervals specified in the table below before use.

| Inspection item | Inspection Interval | Inspection Method |
|---|---|---|
| Cartridge Leakage | Before each procedure, Before every shot during the procedure | Perform a visual inspection prior to use and maintain visual monitoring throughout operation. |
| Cartridge Film Damage | Before each procedure, | Perform a visual inspection prior to use and maintain visual monitoring throughout operation. |
| Cartridge Lifetime | Before every shot during the procedure | Ensure that the cartridge's total shot usage and remaining shot count are verified before each use. |
| Handpiece Button Functionality | Continuously during the procedure | Check that the button switch operates smoothly. |
| External Damage | Before each procedure | Perform a visual inspection. |

## 7.2 Cleaning

1) Main body and connector
   - Use compressed air or similar methods to remove dust from the main unit and connectors.
   - Wipe the outer surfaces of the main unit (front, back, LCD) using a soft cloth lightly moistened with alcohol. Take care to avoid any alcohol entering the electrical terminals or connector contacts.
2) Handpiece
   - To clean the handpiece, disinfect by gently wiping it with a soft cotton pad or cotton swab soaked in ethanol or isopropanol (70–90%) for at least 1 minute.
3) Cartridge
   - Clean the cartridge using a cotton pad moistened with alcohol.

     ※ Always clean and disinfect using only the methods and disinfectants specified above. Do not use any other cleaning agents.

     ※ Do not reuse any disinfecting tools (cotton pads, swabs) once used.

     ※ After cleaning and disinfection, ensure all disinfectant has completely dried before using the device.

## 7.3 Storage

Store the device in a dry, room-temperature environment, away from direct sunlight.

# 8 Troubleshooting

## 8.1 Troubleshooting guidelines

If the issue cannot be resolved using the following methods, contact the manufacturer for service.

| Category | Issue | Recommended action |
|---|---|---|
| Main body | A burning smell or explosion sound is coming from the device | 1) Immediately unplug the power cord from the outlet and contact the manufacturer or authorized distributor. |
| | The device does not power on | 1) Check whether the outlet has power and ensure the plug is securely connected.<br>2) Make sure the device and power cable are properly secured.<br>3) If the device still does not operate after these checks, contact the manufacturer or authorized distributor. |
| | Auxiliary power switch cannot be turned off | 1) Verify that the power cord is connected to the wall outlet and the main power switch is turned on.<br>2) If the issue persists after turning the sub power switch ON/OFF 2–3 times, contact the manufacturer or authorized distributor. |
| | The LCD does not respond to touch | 1) Immediately stop using the device and halt the procedure.<br>2) Turn the sub power switch OFF and then ON again to reboot the system. Check if the issue persists.<br>3) If the problem continues, contact the manufacturer or authorized distributor. |
| | A warning message appears on the LCD | 1) Immediately stop using the device and halt the procedure.<br>2) Refer to “8.2 Warning Code Descriptions” to resolve the issue.<br>3) If the same warning message continues or additional warnings appear, contact the manufacturer or authorized distributor. |
| | System message appears on the LCD | 1) Immediately stop using the device and halt the procedure.<br>2) Refer to “8.2 Error Code Descriptions” to resolve the issue.<br>3) If the same error message continues or additional errors appear, contact the manufacturer or authorized distributor. |
| Cartridge | A continuous rattling or grinding sound comes from the cartridge | 1) Try reinstalling the cartridge. This issue may occur when the cartridge is not properly recognized.<br>2) Replace with a new cartridge and restart the device.<br>3) If the issue persists, contact the manufacturer or authorized distributor. |
| Handpiece | The device powers off when the handpiece shot button is pressed | 1) This may be caused by moisture entering the cartridge. In this case, it is recommended to use a different cartridge.<br>2) If the same issue persists after checking, contact the manufacturer or authorized distributor. |

## 8.2 System messages

### 8.2.1 Caution messages

1) Errors that may occur during device operation on the main screen are classified as either "Caution" or "Error."
2) A caution message is displayed when an issue affecting operation occurs.
3) If a caution code occurs in the READY status, the device will forcibly switch to STANDBY status and display the caution message.

<Caution screen>

※ The caution message (displayed in a red box) can be dismissed by pressing the message.

※ If you switch from STANDBY to READY without resolving the existing caution(s), the remaining caution message(s) will reappear, and the device will not enter the READY status.

| Caution message displayed | Check items | Troubleshooting steps |
|---|---|---|
| Caution<br>High Setting Power<br>Caution Code [UI - 01] | Verify if each cartridge is set above the recommended energy level. | If a caution message appears, press the message check button to acknowledge it. |
| Caution<br>Cartridge Not Connected<br>Caution Code [UI - 02] | Check if the cartridge is properly connected. | 1. Remove and reconnect the cartridge.<br>2. Try connecting another handpiece. |
| Caution<br>Handpiece Not Connected<br>Caution Code [UI - 04] | Check if the handpiece cable is properly connected. | 1. Disconnect and reconnect the handpiece cable.<br>2. Try connecting another handpiece. |
| Caution<br>No Cartridge Shot<br>Caution Code [UI - 08] | Confirm if the cartridge shot count has reached the maximum usage. | Replace with another cartridge. |
| Caution<br>High Temp Sensor Value<br>Caution Code [UI - 10] | Check if the cartridge temperature is high | 1. Wait until the internal cartridge temperature decreases.<br>2. Use another cartridge. |
| Caution<br>UI <-> MAIN Communication Fail<br>Caution Code [UI - 20] | 1. Turn off the power and restart the device. | |
| Caution<br>CTG Forced Removal<br>Caution Code [HP - 0001] | Check if the cartridge was forcibly removed during handpiece operation. | If a caution message appears, press the message check button to acknowledge it. |
| Caution<br>HP Sort Failed<br>Caution Code [HP - 0002] | 1. Remove and reconnect the cartridge.<br>2. Try connecting another handpiece. | |
| Caution<br>P1 Sensor Not Detected<br>Caution Code [HP - 0004] | 1. Remove and reconnect the cartridge.<br>2. Try connecting another cartridge. | |
| Caution<br>P0 Sensor Not Detected<br>Caution Code [HP - 0008] | 1. Remove and reconnect the cartridge.<br>2. Try connecting another cartridge. | |
| Caution<br>RF Output Data Not Corrected<br>Caution Code [HP - 0010] | 1. Remove and reconnect the cartridge.<br>2. Disconnect and reconnect the handpiece cable.<br>3. Turn off the power and restart the device. | |
| Caution<br>Missing Encoder Value<br>Caution Code [HP - 0020] | 1. Remove and reconnect the cartridge.<br>2. Try connecting another handpiece. | |
| Caution<br>Encoder Stop Error<br>Caution Code [HP - 0040] | 1. Remove and reconnect the cartridge.<br>2. Try connecting another handpiece. | |

| Caution message displayed | Check items | Troubleshooting steps |
|---|---|---|
| Caution<br>Motor Time Over<br>Caution Code [HP - 0080] | Check if the handpiece has been used for an extended period. | 1. Remove and reconnect the cartridge.<br>2. Try connecting another handpiece. |
| Caution<br>CTG EEPROM R/W Fail<br>Caution Code [HP - 0100] | 1. Remove and reconnect the cartridge.<br>2. Try connecting another cartridge. | |
| Caution<br>CTG EEPROM Out of Range<br>Caution Code [HP - 0200] | 1. Remove and reconnect the cartridge.<br>2. Try connecting another cartridge. | |
| Caution<br>CTG EEPROM Not Initialized<br>Caution Code [HP - 0400] | 1. Remove and reconnect the cartridge.<br>2. Try connecting another cartridge. | |
| Caution<br>Communication Out of Range<br>Caution Code [HP - 0800] | Check if there are any abnormalities in the treatment data set by the user. | 1. Remove and reconnect the cartridge.<br>2. Turn off the power and restart the device. |
| Caution<br>HP <-> MAIN Communication Fail<br>Caution Code [HP - 1000] | 1. Disconnect and reconnect the handpiece cable.<br>2. Turn off the power and restart the device.<br>3. Try connecting another handpiece. | |
| Caution<br>Data Calculation Error<br>Caution Code [HP - 2000] | 1. Remove and reconnect the cartridge.<br>2. Try connecting another cartridge. | |
| Caution<br>Abnormal Motor Speed<br>Caution Code [HP - 4000] | Check if the handpiece has been used for an extended period. | 1. Remove and reconnect the cartridge.<br>2. Try connecting another handpiece. |
| Caution<br>RF On Time Over<br>Caution Code [MAIN - 0001] | Check if the user-set POWER JOULE time has been exceeded during output. | 1. When a caution message appears, press the message check button.<br>※ If the message occurs consecutively two or more times, please contact the manufacturer immediately. |
| Caution<br>MAIN <-> UI Communication Fail<br>Caution Code [MAIN - 0002] | 1. Turn off the power and restart the device. | |
| Caution<br>MAIN <-> HP Communication Fail<br>Caution Code [MAIN - 0004] | 1. Disconnect and reconnect the handpiece cable.<br>2. Turn off the power and restart the device.<br>3. Try connecting another handpiece. | |

※ If the caution message continues to appear despite taking corrective actions, please contact the manufacturer.

※ [High Setting Power]: When the power is set above the level specified by the manufacturer (Power in Joules), a caution message will be displayed. Pressing the check button acknowledges the message, and the device will operate normally.

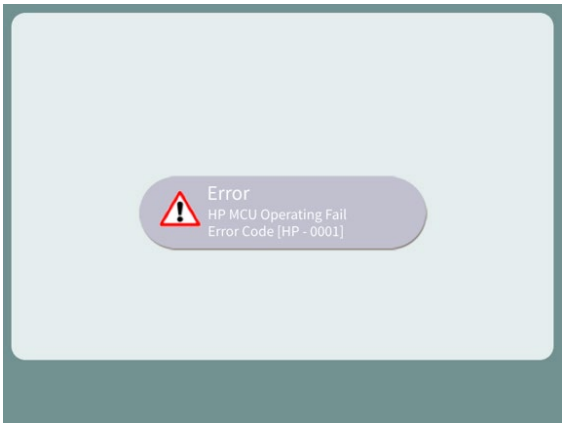
※ [Cartridge Removal and Reinstallation]: When the cartridge is removed from the handpiece and reattached, all existing caution messages are reset and a new inspection begins.

### 8.2.1 Error messages

1) In the event of a critical error, an error message will be displayed.

2) If an error occurs in the READY status, all output will be stopped and the error message will be shown.

3) Once an error message is displayed, the device cannot return to the previous state.




<Error screen>

| Error messages | Troubleshooting steps |
|---|---|
| Error<br>HP MCU Operating Fail<br>Error Code [HP - 0001] | Contact the Manufacturer |
| Error<br>Temperature Over 65'C<br>Error Code [HP - 0002] | Power off the device for approximately one hour, then reboot.<br>Contact the Manufacturer |
| Error<br>MAIN MCU Operating Fail<br>Error Code [MAIN - 0001] | Contact the Manufacturer |
| Error<br>PWR-ADJ Read/Write Fail<br>Error Code [MAIN - 0002] | Contact the Manufacturer |

# 9 IEC Electrical Safety and EMC Specifications

Table 1 IEC Electrical Safety Specifications

| Device Description | |
|---|---|
| This device (LINCURVE PRO (NP-110)) delivers focused ultrasound energy to the lower part of the skin through a focused ultrasound transducer. The irradiated ultrasonic energy forms a small thermal coagulation zone of 1㎟ to 2㎟ in the deep skin layers such as the dermis, the boundary layer between the dermis and subcutaneous fat, the fibrous tissue of the fat layer, the SMAS layer, and the fascia. Tissues are coagulated in the area where ultrasonic energy is accumulated, and gradual thermal damage is also made to nearby tissues, causing various degrees of collagen degeneration. | |
| Mode of Operation: | |
| Electrical Isolation | • Leakage current conforms to IEC 60601-1<br>• Dielectric withstanding voltage conforms to IEC 60601-1 |
| EMC Emissions and Susceptibility: LINCURVE PRO has been tested and found to comply with the limits for medical devices to the IEC 60601-1-2:2014+A1:2020. These limits are designed to provide reasonable protection against harmful interference in a typical medical installation. This Generator generates, uses, and can radiate radiofrequency energy and, if not installed and used in accordance with the instruction given below, may cause harmful interference to other devices in the vicinity. However, there is no guarantee that interference will not occur in a particular installation. | |

Table 2 IEC EMC Specifications (Emissions)

| Guidance and Manufacturer's Declaration – Electromagnetic Emissions | | |
|---|---|---|
| LINCURVE PRO is intended for use in the electromagnetic environment specified below. The customer or the user of LINCURVE PRO should assure that it is used in such an environment. | | |
| Emissions Test | Compliance | Electromagnetic Environment – Guidance |
| RF emissions<br>CISPR 11 | Group 1 | LINCURVE PRO uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment. |
| RF emissions<br>CISPR 11 | Class A | LINCURVE PRO is suitable for use in all establishments other than domestic and those directly connected to the public low voltage power supply network that supplies buildings used for domestic purposes. |
| Harmonic emissions<br>IEC 61000-3-2 | Class A | |
| Voltage fluctuations/ flicker emissions<br>IEC 61000-3-3 | Complies | |

Table 3 IEC EMC Specifications (Immunity)

| Immunity test | Basic EMC standard or test method | IEC 60601 test level | Result |
|---|---|---|---|
| Electrostatic discharge (ESD) | IEC 61000-4-2 | ±8kV Contact<br>±2kV, ±4kV, ±8kV, ±15kV Air | PASS |
| Radio-frequency electromagnetic field. Amplitude modulated | IEC 61000-4-3 | 80 MHz - 2.7 GHz 10 V/m<br>Home Healthcare 10 V/m<br>Prof. Healthcare 3 V/m 80 % / 1 kHz | PASS |
| Power frequency (50 Hz/60 Hz) magnetic field | IEC 61000-4-8 | 30 A/m | PASS |
| Electrical fast transient /burst | IEC 61000-4-4 | 5/50 ns, 100 kHz, ±2 kV | PASS |
| Surge | IEC 61000-4-5 | 1.2/50 (8/20) μs<br>LtL: ±1.0 kV<br>LtG: ±2.0 kV | PASS |
| Conducted RF IEC 61000-4-6 Amplitude modulated | IEC 61000-4-6 | 150 kHz - 80 MHz 3 V<br>ISM Bands 6 V<br>80 % / 1 kHz | PASS |
| Voltage dips, short interruptions and voltage variations on power supply input lines | IEC 61000-4-11 | 0 % $U_T$ for 0.5 cycle (1 phase)<br>0 % $U_T$ for 1 cycle<br>70 % $U_T$ for 25/30 cycles(50/60 Hz)<br>0 % $U_T$ for 250/300 cycles(50/60 Hz) | PASS |

Table 4 IEC Recommended Separation of RF Communication Equipment

| Band (MHz) | Service | IEC 60601 test level | Result |
|---|---|---|---|
| 380–390 | TETRA 400 | 380 - 390 MHz<br>27 V/m; PM 50 %; 18 Hz | PASS |
| 430–470 | GMRS 460,<br>FRS 460 | 430 - 470 MHz<br>28 V/m; (FM ±5 kHz, 1 kHz sine) PM; 18 Hz | PASS |
| 704–787 | LTE Band 13, 17 | 704 - 787 MHz<br>9 V/m; PM 50 %; 217 Hz | PASS |
| 800–960 | GSM 800/900,<br>TETRA 800,<br>iDEN 820,<br>CDMA 850,<br>LTE Band 5 | 800 - 960 MHz<br>28 V/m; PM 50 %; 18 Hz | PASS |
| 1700 –1990 | SM 1800<br>CDMA 1900<br>GSM 1900,<br>DECT,<br>LTE band 1, 3,4, 25, UMTS | 1700 - 1990 MHz<br>28 V/m; PM 50 %; 217 Hz | PASS |
| 2400 –2570 | Bluetooth,<br>WLAN,802.11 b/g/n,<br>RFID2450,<br>LTE Band 7 | 2400 - 2570 MHz<br>28 V/m; PM 50 %; 217 Hz | PASS |
| 5100 –5800 | WLAN 802.11 a/n | 5100 - 5800 MHz<br>9 V/m; PM 50 %; 217 Hz | PASS |

| Rated maximum output power of transmitter P(W) | Minimum separation distance (m) $d = \frac{6}{E}\sqrt{P}$ or 0.3, whichever is greater | | | | | | |
|---|---|---|---|---|---|---|---|
| | 380–390 MHz | 430–470 MHz | 704–787 MHz | 800–960 MHz | 1700 – 1990MHz | 2400 –2570 MHz | 5100 –5800 MHz |
| 0.01 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 |
| 0.1 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 |
| 1 | 0.3 | 0.3 | 0.67 | 0.3 | 0.3 | 0.3 | 0.67 |
| 10 | 0.7 | 0.68 | 2.11 | 0.68 | 0.68 | 0.68 | 2.11 |
| 100 | 2.22 | 2.14 | 6.67 | 2.14 | 2.14 | 2.14 | 6.67 |

# 10 Symbol Descriptions

| Symbol | Description |
|---|---|
| | General prohibition sign |
| | General warning sign |
| | Refer to instruction manual/booklet |
| | "ON" (power) |
| | "OFF" (power) |
| | Stand-by<br>To identify the switch or switch position by means of which part of the equipment is switched on in order to bring it into the stand-by condition.. |
| | Caution |
| CE | CE conformity marking and notified body number |
| EC REP | Authorized representative in the European community |
| SN | Serial number |

| | |
|---|---|
| | Manufacturer |
| | Date of manufacture |
| | WEEE Mark |
| | Temperature limit |
| | This is the correct upright position of the distribution packages for transport and storage. |
| | BF APPLIED PART |
| | Protective earth (ground) |
| | Humidity Limit |
| IPX7 | Cartridge water proof protection class |
| | Do Not Sit Here |
| | No Stepping On Surface |

## 11 Warranty

※ We sincerely thank you for purchasing a product from Newpong Co., Ltd.

※ This product has undergone rigorous quality control and has passed stringent inspections.

※ The compensation standards for repair, exchange, and refund of this product follow the "Consumer Dispute Resolution Standards" announced by the Fair Trade Commission.

※ Should any defects arise under normal usage during the warranty period, Newpong Co., Ltd. will provide complimentary repair services at authorized service centers.

※ In the event of any issues during the warranty period, please promptly notify the company, providing the model name, serial number, date of purchase, and a detailed description of the problem.

※ No part of this user manual may be modified, reproduced, reprinted, or distributed without the prior written consent of Newpong Co., Ltd.

※ Newpong Co., Ltd. reserves the right to amend technical specifications, equipment configurations (including form factors), or manuals without prior notice, in order to enhance product quality and customer satisfaction.

※ The information provided by Newpong Co., Ltd. is accurate and reliable to the best of our knowledge.

※ Newpong Co., Ltd. assumes no liability for damages resulting from the use of the product for purposes other than those outlined in this manual or by unqualified operators.

※ This user manual contains comprehensive information to assist users in properly operating and managing the product. Should you have any inquiries or concerns regarding the use or handling of this device, please contact the manufacturer using the details provided below.

Address: #1216, SJ Technoville 278, Beotkkot-ro, Geumcheon-gu, Seoul, 08511, Republic of Korea

Telephone: +82-2-541-0145

Website: www.newpong.com

※ It is imperative that the user thoroughly reviews this manual prior to using the product to ensure correct operation.

※ This product is classified as a medical device.